

See Beyond: An Integrated AI-Powered Assistive Vision System for Visually Impaired Individuals

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Abstract— This paper presents See Beyond which is an integrated AI-powered assistive platform, designed to enhance the independence, safety, and quality of life of visually impaired individuals. Many critical challenges such as obstacle detection, text recognition, people identification, and currency classification are addressed through a coordinated hardware-software co-design approach. The hardware is built around a cane with distance sensor, and power management unit. The software comprises four core modules: (1) a YOLO-based real-time object detection system for obstacle and object awareness, (2) a Paddle OCR-based scene text recognition module for reading signboards, labels, and printed material, (3) an embedding-based face recognition module using Face Net for identifying known individuals without internet connectivity, and (4) a YOLOv8-based Pakistani currency detection module achieving 97.56% accuracy on a custom dataset. A Flutter-based companion mobile application supports system configuration and secure caregiver management. Experimental results confirm that all core modules operate reliably in real-world indoor and outdoor conditions. Our system demonstrates that affordable embedded AI can meaningfully close the gap in comprehensive assistive technology.

Assistive technology, visually impaired, object detection, YOLOv8, face recognition, FaceNet, OCR, PaddleOCR, FastAPI, Flutter, text-to-speech, speech-to-text, voice-controlled interface, ultrasonic sensors, haptic feedback, edge AI, TensorFlow Lite, cosine similarity, RESTful API.

I. INTRODUCTION

This template, modified in MS Word 2007 and saved as a “Word 97-2003 Document” for the PC, provides authors with most of the formatting specifications needed for preparing electronic versions of their papers. All standard paper components have been specified for three reasons: (1) ease of use when formatting individual papers, (2) automatic compliance to electronic requirements that facilitate the concurrent or later production of electronic products, and (3) conformity of style throughout a conference proceedings. Margins, column widths, line spacing, and type styles are built-in; examples of the type styles are provided throughout this document and are identified in italic type, within parentheses, following the example. Some components, such as multi-leveled equations, graphics, and tables are not prescribed, although the various table text styles are provided.

The formatter will need to create these components, incorporating the applicable criteria that follow.

II. SYSTEM ARCHITECTURE

The system follows a hardware-software co-design with six development phases: requirement analysis, hardware selection, ML model development, mobile app development, integration/testing, and optimization. The hardware centers on a smart cane with dual ultrasonic sensors (near-range and far-range) and an Arduino Nano controller driving a vibration motor through a transistor circuit with flyback protection. The software layer comprises a FastAPI backend and a Flutter frontend communicating over HTTP multipart requests.

III. PERCEPTION MODULES

A. Object Detection:

A YOLOv8 model fine-tuned on 77 custom classes relevant to daily navigation (doors, stairs, vehicles, furniture) achieved precision of 63.9%, recall of 48.3%, and mAP@50 of 52.5% over 80 epochs.

B. Currency Detection:

A custom dataset of 400 real-world images of Pakistani banknotes (seven denominations) trained with YOLOv8s over 65 epochs achieved 97.56% final accuracy, with precision 0.94, recall 0.793, and mAP@50 of 0.96.

C. OCR:

Three approaches were evaluated, Google ML Kit, YOLOv8-based text localization, and PaddleOCR. PaddleOCR's end-to-end DB-detection and CRNN-CTC recognition pipeline delivered the highest accuracy (confidence 0.84–1.00) on scene text but required further optimization for mobile deployment; EasyOCR was adopted in the deployed backend for unconstrained text recognition.

D. Face Recognition:

A four-stage pipeline, RetinaFace detection, geometric alignment, FaceNet (quantized TFLite, 512-dimensional embeddings), and cosine similarity matching against a local

JSON database enables privacy-preserving, scalable enrollment without retraining.

IV. BACKEND ARCHITECTURE (FASTAPI)

The backend is implemented in Python using FastAPI for asynchronous request handling. A layered design separates the API routing layer, service layer, and model inference layer (RetinaFace, FaceNet, EasyOCR, YOLOv8). A dependency-injection container loads all models once at startup (singleton pattern) to avoid repeated disk I/O. Pydantic-based configuration management and exception handling ensure robust, typed, and secure operation. REST endpoints (/api/object/detect, /api/currency/analyze, /api/ocr/read, /api/face/analyze, /api/face/register, /api/blind-assist/analyze) return structured JSON responses consumed by the mobile client. Performance optimizations include model quantization, vectorized NumPy similarity computation, and non-blocking asynchronous I/O.

V. MOBILE APPLICATION (FLUTTER)

The Flutter frontend provides a hands-free, voice-driven interface built around five subsystems: speech-to-text wake-word and command detection, intent recognition via keyword matching, a shared camera provider with native Android foreground-service fallback for locked-screen capture, REST communication with the FastAPI backend, and text-to-speech output. A continuous asynchronous voice loop cycles through resting, command-listening, processing, and speaking states, enabling fully hands-free operation for reading text, detecting objects, identifying currency, recognizing faces, and describing scenes.

VI. SMART CANE HARDWARE

The cane integrates two HC-SR04 ultrasonic sensors (near-range ~20 cm, far-range ~60 cm) controlled by an Arduino Nano. Obstacles between 20–40 cm trigger pulsed haptic feedback as an early warning, while the motor remains off beyond 40 cm to conserve battery. An NPN transistor driver with a flyback diode safely controls the vibration motor.

VII. RESULTS AND DISCUSSION

Prototype testing across indoor and outdoor environments demonstrated consistent operation of all modules. Currency detection achieved 97.56% accuracy on 82 test samples. PaddleOCR delivered high-confidence scene text recognition with minor errors on stylized/occluded text. The face recognition module reliably identified pre-registered individuals with scalable enrollment. The object detection module provided timely audio alerts, and the voice-activated Flutter interface ensured hands-free operation throughout.

VIII. CONCLUSION AND FUTURE WORK

See Beyond demonstrates that an affordable, modular AI-powered assistive system combining a FastAPI backend, Flutter frontend, and a smart cane can meaningfully improve independence for visually impaired users. Future work includes embedded deployment (e.g., ESP32), GPS/BLE/SLAM-based navigation, dataset expansion, full on-device PaddleOCR optimization, and extensive user testing with visually impaired participants.

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